

Assignment-5

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1. Flatten Binary Tree to Linked List

Python

```
class Solution:
    def flatten(self, root: TreeNode) -> None:
        curr = root

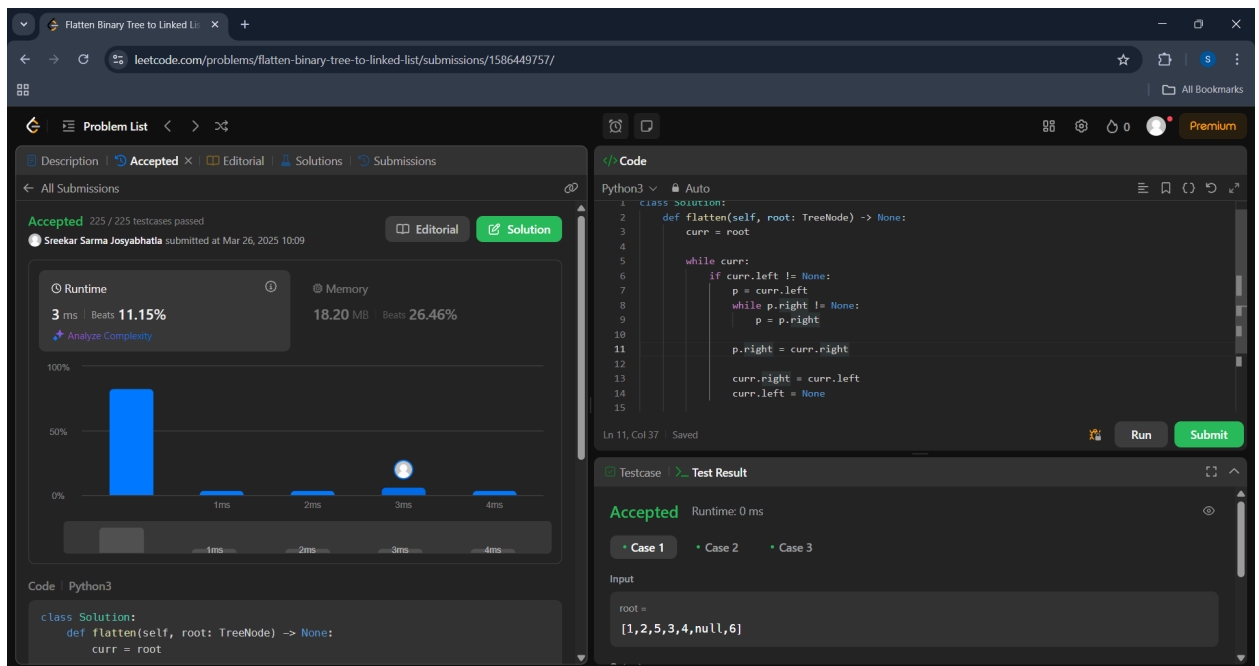
        while curr:
            if curr.left != None:
                p = curr.left
                while p.right != None:
                    p = p.right

                p.right = curr.right

            curr.right = curr.left
            curr.left = None

            curr = curr.right
```

OUTPUT



2. Binary Tree Zigzag Level Order Traversal

Python

```
class Solution:
    def zigzagLevelOrder(self, root: Optional[TreeNode]) -> List[List[int]]:

        if not root:
            return []
        result = []
        q = deque([root])
        ltr = True
        while q:
            size = len(q)
            temp = [0] * size
            for i in range(size):
                node = q.popleft()
                index = i if ltr else (size - 1 - i)
                temp[index] = node.val
                if node.left:
                    q.append(node.left)
                if node.right:
                    q.append(node.right)
            ltr = not ltr
            result.append(temp)
        return result
```

OUTPUT

The screenshot shows a LeetCode submission for the problem "Binary Tree Zigzag Level Order Traversal II". The submission is accepted, with a runtime of 0 ms and memory usage of 18.15 MB. The code is in Python3 and implements a recursive solution for zigzag level order traversal.

```

17  temp = []
18  for i in range(size):
19      node = q.popleft()
20      index = i if ltr else (size - 1 - i)
21      temp[index] = node.val
22      if node.left:
23          q.append(node.left)
24      if node.right:
25          q.append(node.right)
26      ltr = not ltr
27      result.append(temp)
28  return result
29

```

3. Binary Tree Level Order Traversal II

Python

class Solution:

```

def levelOrderBottom(self, root: Optional[TreeNode]) -> List[List[int]]:
    if not root:
        return []
    qu = deque([])
    qu.append(root)
    res = []
    while qu:
        temp = []
        for _ in range(len(qu)):
            node = qu.popleft()
            temp.append(node.val)
            if node.left:
                qu.append(node.left)
            if node.right:
                qu.append(node.right)
        res.append(temp)
    return res[::-1]

```

OUTPUT

Binary Tree Level Order Traversal II

Accepted 34 / 34 testcases passed

Sreekar Sarma Josyabhatla submitted at Mar 26, 2025 10:16

Runtime: 4 ms | Beats: 3.71% | Memory: 18.20 MB | Beats: 17.02%

Code: Python3

```
class Solution:
    def levelOrderBottom(self, root: Optional[TreeNode]) -> List[List[int]]:
        if not root:
            return []
        qu = [root]
        res = []
        while qu:
            temp = []
            for _ in range(len(qu)):
                node = qu.popleft()
                temp.append(node.val)
                if node.left:
                    qu.append(node.left)
                if node.right:
                    qu.append(node.right)
            res.append(temp)
        return res[::-1]
```

Testcase: Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input: root = [3, 9, 20, null, null, 15, 7]

4. Sum Root to Leaf Numbers

Python

class Solution:

```
def sumNumbers(self, root: Optional[TreeNode]) -> int:
    tot_sum, cur, depth = 0, 0, 0
    while root:
        if root.left:
            pre, depth = root.left, 1
            while pre.right and pre.right != root:
                pre, depth = pre.right, depth + 1
            if not pre.right:
                pre.right = root
                cur = cur * 10 + root.val
                root = root.left
            else:
                pre.right = None
                if not pre.left: tot_sum += cur
                cur //= 10**depth
                root = root.right
        else:
            cur = cur * 10 + root.val
            if not root.right: tot_sum += cur
```

```

        root = root.right
    return tot_sum

```

OUTPUT

The screenshot shows a LeetCode submission for the problem "Sum Root to Leaf Numbers". The submission is accepted, with a runtime of 0 ms and memory usage of 17.85 MB. The code is in Python3 and implements a recursive function to calculate the sum of root-to-leaf numbers. The test case input is [1,2,3] and the output is 1.

```

class Solution:
    def sumNumbers(self, root: Optional[TreeNode]) -> int:
        tot_sum, cur, depth = 0, 0, 0
        while root:
            if root.left:
                pre, depth = root.left, 1
                while pre.right and pre.right != root:
                    pre, depth = pre.right, depth + 1
                if not pre.right:
                    pre.right = root
                    cur = cur * 10 + root.val
                    root = root.left
            else:
                pre.right = None

```

5. Vlad and Unfinished Business

C/C++

```

#include <bits/stdc++.h>

using namespace std;

// #define int long long
// #define ll long long
// #define double long double
#define all(a) a.begin(), a.end()
#define rall(a) a.rbegin(), a.rend()

const int MOD = 1e9 + 7;
const int maxN = 5e3 + 1;
const int INF = 2e9;

```

```

const int MB = 20;

vector<vector<int>> g;
vector<bool> todo, good;

void dfs(int v, int p = -1) {
    for (int u : g[v]) {
        if (u != p) {
            dfs(u, v);
            if (todo[u]) {
                todo[v] = true;
            }
            if (good[u]) {
                good[v] = true;
            }
        }
    }
}

void solve() {
    int n, k;
    cin >> n >> k;
    g.clear();
    g.resize(n);
    int x, y;
    cin >> x >> y;
    --x;
    --y;
    todo.resize(n);
    fill(all(todo), false);
    good.resize(n);
    fill(all(good), false);
    for (int i = 0; i < k; ++i) {
        int v;
        cin >> v;
        --v;
        todo[v] = true;
    }
    good[y] = true;
    for (int i = 0; i < n - 1; ++i) {

```

```

        int v, u;
        cin >> v >> u;
        --v;
        --u;
        g[v].push_back(u);
        g[u].push_back(v);
    }
    dfs(x);
    int ans = 0;
    for (int i = 0; i < n; ++i) {
        if (i == x) {
            continue;
        }
        if (good[i]) {
            ++ans;
        } else if (todo[i]) {
            ans += 2;
        }
    }
    cout << ans << '\n';
}

signed main() {
    ios_base::sync_with_stdio(false);
    cin.tie(0);
    cout.tie(0);
    // srand(time(0));
    int t = 1;
    cin >> t;
    while (t--) solve();
}

```

OUTPUT:

The screenshot shows the Codeforces website interface. At the top, there's a navigation bar with links like HOME, TOP, CATALOG, CONTESTS, GYM, PROBLEMSET, GROUPS, RATING, EDU, API, CALENDAR, HELP, and RAYAN. Below this, there's a section for 'My Submissions' with a table listing recent submissions. The table has columns for #, When, Who, Problem, Lang, Verdict, Time, and Memory. The submissions listed are for the problem 'F. Vlad and Unfinished Business' by user 'sreekarsarma2003'. The first submission (312520947) was accepted in C++17 (GCC 7-32) with a runtime of 296 ms and memory of 14600 KB. The second (312520851) had a runtime error. The third (312520684) had a wrong answer. The fourth (312520455) had a wrong answer. To the right of the table, there's a sidebar with a 'Codeforces Round 787 (Div. 3)' section, a 'Finished' status, a 'Practice' button, and a 'Virtual participation' section with a 'Start virtual contest' button. At the bottom of the sidebar, there's a 'Clone Contest to Mashup' section with a 'Clone Contest' button.

#	When	Who	Problem	Lang	Verdict	Time	Memory
312520947	Mar/26/2025 10:33:19.3	sreekarsarma2003	F. Vlad and Unfinished Business	C++17 (GCC 7-32)	Accepted	296 ms	14600 KB
312520851	Mar/26/2025 10:32:19.3	sreekarsarma2003	F. Vlad and Unfinished Business	Python 3	Runtime error on test 1	78 ms	2000 KB
312520684	Mar/26/2025 10:30:19.3	sreekarsarma2003	F. Vlad and Unfinished Business	Python 3	Wrong answer on test 1	77 ms	100 KB
312520455	Mar/26/2025 10:27:19.3	sreekarsarma2003	F. Vlad and Unfinished Business	Python 3	Wrong answer on test 1	15 ms	200 KB

6. G. Remove Directed Edges

C/C++

```
#include <bits/stdc++.h>
```

```
#define forn(i, n) for (int i = 0; i < int(n); i++)
```

```
using namespace std;
```

```
const int INF = 1e9;
```

```
vector<int> in, out;
vector<vector<int>> g;
vector<int> dp;
```

```
int calc(int v){
    if (dp[v] != -1) return dp[v];
    if (in[v] >= 2 && out[v] >= 2){
        dp[v] = 1;
        for (int u : g[v])
            dp[v] = max(dp[v], calc(u) + 1);
    }
    else if (in[v] >= 2){
        dp[v] = 1;
    }
    else{
```



```

        dp[v] = -INF;
    }
    return dp[v];
}

int main() {
    int n, m;
    scanf("%d%d", &n, &m);
    g.resize(n);
    in.resize(n);
    out.resize(n);
    forn(i, m){
        int v, u;
        scanf("%d%d", &v, &u);
        --v, --u;
        g[v].push_back(u);
        ++in[u];
        ++out[v];
    }
    int ans = 1;
    dp.resize(n, -1);
    forn(v, n) if (out[v] >= 2){
        for (int u : g[v]){
            ans = max(ans, calc(u) + 1);
        }
    }
    printf("%d\n", ans);
    return 0;
}

```

OUTPUT

312521384	Mar/26/2025 10:38UTC+5.5	sreekarsama2003	1674G - Remove Directed Edges	C++17 (GCC 7-32)	Accepted	218 ms	3200 KB
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7. Fall Down

C/C++

```

#include <iostream>
#include <vector>

```

```

int main(){

    std::ios_base::sync_with_stdio(false);
    long t; std::cin >> t;
    while(t--){
        long nr, nc; std::cin >> nr >> nc;
        std::vector<std::string> v(nr);
        for(long row = 0; row < nr; row++){std::cin >> v[row];}

        for(long col = 0; col < nc; col++){
            long b(nr - 1);
            for(long row = nr - 1; row >= 0; row--){
                if(v[row][col] == '*'){
                    v[row][col] = '.';
                    v[b][col] = '*';
                    --b;
                }
                else if(v[row][col] == 'o'){b = row - 1;}
            }
        }

        for(long row = 0; row < nr; row++){std::cout << v[row] <<
std::endl;}
    }
}

```

Output:

312521569	Mar/26/2025 10:41 UTC+5.5	sreekarsarma2003	1669G - Fall Down	C++17 (GCC 7-32)	Accepted	62 ms	0 KB
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8.1670C - Where is the Pizza?

C/C++

```

#include <bits/stdc++.h>

using namespace std;
const int N = 100100;
int n;

```

```

int a[N], b[N], c[N];
bool used[N];
int par[N], sz[N];

int getPar(int v) {
    return par[v] == v ? v : (par[v] = getPar(par[v]));
}

bool unite(int v, int u) {
    v = getPar(v);
    u = getPar(u);
    if (v == u) return false;
    if (sz[v] < sz[u]) swap(v, u);
    sz[v] += sz[u];
    par[u] = v;
    return true;
}

void solve() {
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        used[i] = 0;
        par[i] = i;
        sz[i] = 1;
    }
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);
    for (int i = 0; i < n; i++) {
        scanf("%d", &c[i]);
        if (c[i] != 0) used[c[i]] = 1;
    }
    long long ans = 1, mod = 1e9+7;
    for (int i = 0; i < n; i++) {
        if (c[i] != 0) continue;
        int v = a[i], u = b[i];
        if (used[v] || used[u] || v == u) continue;
        if (!unite(v, u)) ans = (ans*2)%mod;
    }
    printf("%u\n", ans);
}

int main()

```

```
{  
  
    int t;  
    scanf("%d", &t);  
    while(t--) solve();  
  
    return 0;  
}
```

Output:

312522057	Mar/26/2025 10:46 ^{UTC+5.5}	sreekarsarma2003	1670C - Where is the Pizza?	C++17 (GCC 7-32)	Accepted	124 ms	2100 KB
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